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Analysis of wave interaction with submerged perforated semi-circular breakwaters through multipole method

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ABSTRACT

This study gives a solution of water wave interaction with submerged perforated semi-circular breakwaters in the context of linear potential theory. The multipole expansion with singularities located on the seabed is developed for the solution. It is found that the expanded series converges very rapidly as the number of terms increase. The wave reflection and transmission coefficients, the horizontal and vertical wave forces and the hydrodynamic pressure distribution over the semi-circular breakwaters are obtained and the results are carefully examined. Their significance for practical engineering is discussed.

1. Introduction

Perforated structures have been often used in coastal and offshore engineering [1]. One of important perforated structures is the semi-circular caisson breakwater, which was initially built at Miyazaki Port, Japan in early 1990s [2]. This type of precast reinforced concrete caisson structure mainly includes two components: an impermeable or perforated semi-circular arc and a bottom slab [3]. The major merits of semi-circular breakwaters are their high stability against sliding, zero overturning moment acting on the caisson as the hydrodynamic pressure acting on the structure surface passes through the centre of the circle, light weight which is good for soft soil foundation, and easy for construction and removal [4]. Due to these merits, the semi-circular structures have been often used in practical engineering. For example in China, about 527 miles of semi-circular breakwater was built at Tianjin Port in 1997, and about 18 km of submerged semi-circular caisson jetties was constructed for the Deep Channel Improvement Project of the Yangtze River Estuary [5].

The semi-circular breakwaters may be submerged or surface piercing corresponding to different design water levels [3]. Tanimoto and Takahashi [6] developed an empirical formula to calculate the wave forces acting on the semi-circular breakwaters by introducing the phase and angle modification coefficients into the Goda's formula [7] for vertical walls. Based on the experimental data, Xie [4] found that the method of Tanimoto and Takahashi [6] may not be suitable for the submerged semi-circular breakwaters and they gave another empirical formula for the wave forces acting on the submerged semi-circular breakwaters. In a series of studies, Dhinakaran et al. [8–10] examined the effects of the rubble mound foundation and the porosity of caisson face on the hydrodynamic performance of semi-circular breakwaters. Zhang et al. [5] considered obliquely incident waves and obtained the effect of incident angle on wave forces acting on the semi-circular breakwater.

Numerical studies on semi-circular structures have also been carried out by many researchers, principally based on the boundary element method. Forbes and Schwartz [11] and Forbes [12] considered the uniform incoming stream over a semi-circular obstacle at supercritical, subcritical and critical conditions defined through the Froude number with the water depth. Cooker et al. [13] simulated the interaction of solitary waves with a submerged semi-circular obstacle and compared their numerical results with the experimental data. Jia [14] examined the interaction between Stokes waves and semi-circular breakwaters. Yuan and Tao [3] used the boundary element method for the governing Laplace equation and the finite difference method for the free surface boundary in the time domain to simulate wave interaction with semi-circular breakwaters. Based on the deep submergence assumption, Chakrabarti and Naftzger [15] obtained an approximate solution for wave motion over a submerged semi-circular cylinder. Very recently, Kasem and Sasaki [16] simulated the propagation of water waves over a semi-circular breakwater by solving the incompressible Navier–Stokes equations. While the above work is on rigid structures, Dewia et al. [17] considered a submerged impermeable flexible semi-circular

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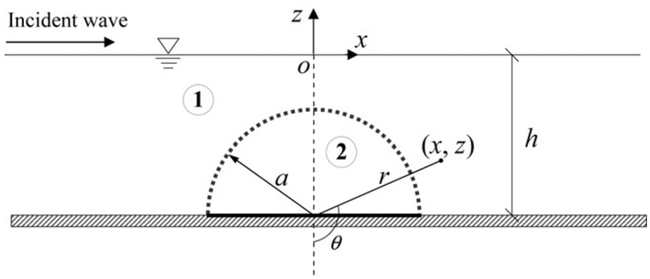

Fig. 1. Sketch for wave interaction with a submerged semi-circular breakwater.

shell filled with water. All these studies are for impermeable semi-circular breakwaters and there are far fewer numerical or analytical-oriented studies on perforated cases.

To give a better understanding of the hydrodynamic performance of perforated semi-circular breakwaters, the present study will develop a semi-analytical solution for wave interaction with submerged perforated semi-circular breakwaters by using the multipole expansion method. The multipole expansion was first used, in water wave and body problems, by Ursell [18] to study the heaving motion of a half-immersed circular cylinder in deep water. More studies based on the multipole expansion can be found in [19–27]. In the following section, the boundary problem of wave interaction with a submerged perforated semi-circular breakwater is formulated. In Section 3, the solution of wave interaction with a submerged perforated semi-circular breakwater is developed by means of the multipole expansions. In Section 4, the special case of a submerged impermeable semi-circular breakwater is examined. In Section 5, numerical examples are given to examine the hydrodynamic performance of breakwaters. Finally the main conclusions are drawn.

2. Governing equation

The interaction of incident regular waves with a submerged semi-circular breakwater which consists of a thin perforated semi-circular arc and a bottom slab is sketched in Fig. 1. A Cartesian coordinate (x, z), with the origin at the still water level, the z-axis measured vertically upwards, is used to describe the 2D problem. Also a polar coordinate system is defined with $r\cos\theta = -(z+h)$ and $r\sin\theta = x$. Here the water depth h is assumed to be constant. The incident wave propagates along the positive x-direction. The radius of the semi-circular breakwater is a $(a < h)$, and its centre is located at $(0, -h)$. For the convenience, the outer fluid domain of the breakwater is named as region 1, and the inner fluid domain as region 2.

Based on assumptions of inviscid liquid and irrotational flow, the fluid motion due to a regular incoming wave can be described by a velocity potential:

$$\Phi(x, z, t) = \mathrm{Re}[\phi(x, z)\mathrm{e}^{-\mathrm{i}\omega t}], \tag{1}$$

where Re denotes the real part of the function, $\phi(x, z)$ is the spatial velocity potential; $\mathrm{i} = \sqrt{-1}$; ω is the frequency of the incoming wave; and t is the time. When the liquid is incompressible, the velocity potential satisfies the Laplace equation:

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial z^2} = 0. \tag{2}$$

For the linear problem, the velocity potential also satisfies the following free surface boundary conditions:

$$\frac{\partial\phi}{\partial z} = K\phi, \quad z = 0, \tag{3}$$

where $K = \omega^2/g$ and g is the gravitational acceleration. On the rigid surface of the sea bed or the bottom slab, we have

$$\frac{\partial\phi}{\partial z} = 0, \quad z = -h. \tag{4}$$

In the far fields, we have when the incoming wave is from the left hand side:

$$\lim_{x\to\pm\infty}\left(\frac{\partial}{\partial x} \mp \mathrm{i}k\right)\begin{pmatrix}\phi \\ \phi - \phi^{\mathrm{I}}\end{pmatrix} = 0, \tag{5}$$

where k is the wave number and satisfies:

$$K = k\tanh kh, \tag{6}$$

and ϕ^{I} is the incident potential and can be written as:

$$\phi^{\mathrm{I}} = -\frac{\mathrm{i}gH}{2\omega}\frac{\cosh k(z+h)}{\cosh kh}\mathrm{e}^{\mathrm{i}kx}, \tag{7}$$

in which H is the height of the incident wave.

On the surface of the perforated semi-circular arc, we have [28]:

$$\frac{\partial\phi_1}{\partial r} = \frac{\partial\phi_2}{\partial r} = \mathrm{i}kG(\phi_2 - \phi_1), \quad r = a, \tag{8}$$

where the subscripts 1 and 2 denote velocity potentials in regions 1 and 2, respectively; and G is the porous effect parameter [28]:

$$G = \frac{\varepsilon}{k\delta(f - \mathrm{i}s)}, \tag{9}$$

in which, δ is the plate thickness; and ε, f and s are, respectively, the geometrical porosity, the resistance coefficient and the inertial effect coefficient of the perforated plate. We note that when G is zero, the semi-circular arc becomes impermeable. When the absolute value of G tends to infinity, the semi-circular arc disappears. We also note that the first equal sign in Eq. (8) indicates the continuity of normal fluxes at the perforated plate. The second equal sign indicates that the normal fluid velocity passing through the perforated plate is linearly proportional to the pressure difference between the both sides of the arc. A different quadratic relationship between the pressure difference and the normal fluid velocity across the porous plate can be found in Molin [1].

3. The multipole expansion

We use the multipole expansion method to solve the above problem. The detailed derivation of the expansion is given in Appendix A.

Using the Taylor expansion of the exponential function, the incident potential in Eq. (7) can be written as:

$$\phi^{\mathrm{I}} = \frac{-\mathrm{i}gH}{2\omega}\frac{1}{\cosh\,kh}\left[\sum_{n=0}^{\infty}\frac{(kr)^{2n}}{(2n)!}\cos 2n\theta + \mathrm{i}\sum_{n=1}^{\infty}\frac{(kr)^{2n-1}}{(2n-1)!}\sin(2n-1)\theta\right]. \tag{10}$$

The two summations in the square brackets of the above equation are symmetric and antisymmetric about z axis respectively. Correspondingly, we split ϕ_1 and ϕ_2 into a symmetric component ϕ^S and an antisymmetric component ϕ^A:

$$\phi_j = \phi_j^S + \phi_j^A, \quad j = 1, 2. \tag{11}$$

Based on the derivations in Appendix A, the two components of ϕ_1 can be written as:

$$\phi_1^S = \frac{-\mathrm{i}gH}{2\omega}\frac{1}{\cosh\,kh}\left[\sum_{n=0}^{\infty}\frac{(kr)^{2n}}{(2n)!}\cos 2n\theta + \sum_{n=1}^{\infty}a^{2n}c_n^S\varphi_n^S\right], \tag{12}$$

$$\phi_1^A = \frac{-\mathrm{i}gH}{2\omega}\frac{1}{\cosh\ kh}\left[\mathrm{i}\sum_{n=1}^{\infty}\frac{(kr)^{2n-1}}{(2n-1)!}\ \sin(2n-1)\theta + \sum_{n=1}^{\infty}a^{2n-1}c_n^A\varphi_n^A\right], \tag{13}$$

where c_n^S and c_n^A are unknown expansion coefficients; and the multipoles φ_n^S and φ_n^A are given in Eqs. (A.13) and (A.14). By the separation of variables method, the two components of ϕ_2 satisfying the Laplace equation and the water bottom condition, Eq. (4), can be written as:

$$\phi_2^S = \frac{-\mathrm{i}gH}{2\omega}\frac{1}{\cosh\ kh}\left[d_0^S + \sum_{n=1}^{\infty}d_n^S r^{2n}\cos 2n\theta\right], \tag{14}$$

$$\phi_2^A = \frac{-\mathrm{i}gH}{2\omega}\frac{1}{\cosh\ kh}\left[\sum_{n=1}^{\infty}d_n^A r^{(2n-1)}\sin(2n-1)\theta\right], \tag{15}$$

where d_n^S and d_n^A are unknown coefficients.

Inserting the expressions of ϕ_1 and ϕ_2 into the porous boundary conditions in Eq. (8), multiplying both sides of the new equations by $\cos 2m\theta$ and $\sin(2m-1)\theta$, respectively, then integrating results with respect to θ from $\pi/2$ to $3\pi/2$ and using the orthogonality of trigonometric functions, we have:

$$c_m^S - \sum_{n=1}^{\infty}a^{2(m+n)}C_{mn}^S c_n^S + a^{2m}d_m^S = \frac{(ka)^{2m}}{(2m)!}, \quad m = 1, 2, 3, \ldots, \tag{16}$$

$$\sum_{n=1}^{\infty}a^{2n}C_{0n}^S c_n^S - d_0^S = -1, \tag{17a}$$

$$c_m^S + \sum_{n=1}^{\infty}a^{2(m+n)}C_{mn}^S c_n^S + a^{(2m-1)}\left(\frac{2m}{\mathrm{i}kG} - a\right)d_m^S = -\frac{(ka)^{2m}}{(2m)!}, \quad m = 1, 2, 3, \ldots, \tag{17b}$$

$$c_m^A - \sum_{n=1}^{\infty}a^{2(m+n-1)}C_{mn}^A c_n^A + a^{2m-1}d_m^A = \frac{\mathrm{i}(ka)^{2m-1}}{(2m-1)!}, \quad m = 1, 2, 3, \ldots, \tag{18}$$

$$c_m^A + \sum_{n=1}^{\infty}a^{2(m+n-1)}C_{mn}^A c_n^A + a^{2(m-1)}\left(\frac{2m-1}{\mathrm{i}kG} - a\right)d_m^A = -\frac{\mathrm{i}(ka)^{2m-1}}{(2m-1)!}, \quad m = 1, 2, 3, \ldots, \tag{19}$$

where C_{mn}^S and C_{mn}^A are given in Eqs. (A.15) and (A.16). Based on the desired accuracy, we can solve the above four sets of linear systems by truncating m and n at M. Then the obtained values of expansion coefficients, c_m^S, c_m^A, d_m^S and d_m^A can be used to find various parameters of interest.

At the far fields, we have [29]:

$$\phi_1 = -\frac{\mathrm{i}gH}{2\omega}(e^{\mathrm{i}kx} + Re^{-\mathrm{i}kx})\frac{\cosh k(z+h)}{\cosh kh}, \quad x \to -\infty, \tag{20}$$

$$\phi_1 = -\frac{\mathrm{i}gH}{2\omega}T\frac{\cosh\ k(z+h)}{\cosh kh}e^{\mathrm{i}kx}, \quad x \to +\infty. \tag{21}$$

where R and T are the reflection and transmission coefficients respectively. Then according to Eqs. (7), (A.10) and (A.12), the magnitudes of the reflection and transmission coefficients can be calculated by

$$C_R = \left|R\right| = \left|\sum_{n=1}^{M}a^{2n}c_n^S\frac{\mathrm{i}\pi k^{2n-1}}{2hN_0{}^2(2n-1)!} - \sum_{n=1}^{M}a^{2n-1}c_n^A\frac{\pi k^{2n-2}}{2hN_0{}^2(2n-2)!}\right|, \tag{22}$$

$$C_T = \left|T\right| = \left|1 + \sum_{n=1}^{M}a^{2n}c_n^S\frac{\mathrm{i}\pi k^{2n-1}}{2hN_0{}^2(2n-1)!} + \sum_{n=1}^{M}a^{2n-1}c_n^A\frac{\pi k^{2n-2}}{2hN_0{}^2(2n-2)!}\right|, \tag{23}$$

When waves pass through perforated structures, the wave energy dissipation occurs, and the energy loss coefficient is defined as:

$$C_L = 1 - C_R^2 - C_T^2. \tag{24}$$

There is no energy loss at $C_L = 0$. However, the incident wave energy is totally dissipated at $C_L = 1$.

The dynamic pressure $p(x, z)$ can be obtained by the linear Bernoulli equation:

$$p(x, z) = \mathrm{i}\rho\omega\phi(x, z), \tag{25}$$

where ρ is the fluid density. The wave force acting on the breakwater can be obtained by integrating the dynamic pressure along the structure surface. The total horizontal wave force acting on the breakwater can be calculated by:

$$F_x = \mathrm{i}\rho\omega\int_{\pi/2}^{3\pi/2}[\phi_2(a,\theta) - \phi_1(a,\theta)]a\sin\theta\,d\theta = \frac{\rho\omega}{kG}\int_{\pi/2}^{3\pi/2}\left.\frac{\partial\phi_2^A(r,\theta)}{\partial r}\right|_{r=a}a\ \sin\theta\,d\theta = \frac{\pi\rho gHa}{4\mathrm{i}kG\cosh kh}d_1^A, \tag{26}$$

where we have use the porous boundary condition in Eq. (8). The total vertical wave force acting on the breakwater can be calculated by:

$$F_z = F_{z1} + F_{z2}, \tag{27}$$

where F_{z1} and F_{z2} are vertical wave forces acting on the perforated semi-circular arc and the bottom slab:

$$F_{z1} = \mathrm{i}\rho\omega\int_{\pi/2}^{3\pi/2}[\phi_2(a,\theta) - \phi_1(a,\theta)]a(-\cos\theta)\,d\theta = \frac{\rho\omega}{kG}\int_{\pi/2}^{3\pi/2}\left.\frac{\partial\phi_2^S(r,\theta)}{\partial r}\right|_{r=a}a(-\cos\ \theta)\,d\theta = \frac{-2\rho gHa}{\mathrm{i}kG\cosh kh}\sum_{n=1}^{M}\frac{(-1)^n n d_n^S a^{2n-1}}{4n^2-1}, \tag{28}$$

$$F_{z2} = -2\mathrm{i}\rho\omega\int_0^a\phi_2^S(r,\pi/2)\ dr = \frac{-\rho gHa}{\cosh kh}\sum_{n=0}^{M}\frac{(-1)^n a^{2n}}{2n+1}d_n^S. \tag{29}$$

In Eq. (28), the porous boundary condition in Eq. (8) has been used.

4. The special case of an impermeable breakwater

For the special case of a submerged impermeable semi-circular breakwater with $G = 0$, we only need consider the velocity potential

ϕ_1 in Eqs. (12) and (13). Then the porous boundary condition in Eq. (8) becomes:

$$\frac{\partial \phi_1}{\partial r} = 0, \quad r = a, \tag{30}$$

and Eqs. (16)–(19) become:

$$c_m^S - \sum_{n=1}^{\infty} a^{2(m+n)} C_{mn}^S c_n^S = \frac{(ka)^{2m}}{(2m)!}, \quad m = 1, 2, 3, \ldots, \tag{31}$$

$$c_m^A - \sum_{n=1}^{\infty} a^{2(m+n-1)} C_{mn}^A c_n^A = \frac{\mathrm{i}(ka)^{2m-1}}{(2m-1)!}, \quad m = 1, 2, 3, \ldots, \tag{32}$$

Solving Eqs. (31) and (32) give the expansion coefficients in the velocity potential ϕ_1. Then the magnitudes of the reflection and transmission coefficients are still calculated by Eqs. (22) and (23), respectively. However the energy loss coefficient for the impermeable case is always equal to zero, which is in fact a well known mathematical identity based on the energy conservation. The total horizontal and vertical wave forces acting on the impermeable breakwater can be calculated by:

$$\begin{aligned} F_x &= \mathrm{i}\rho\omega \int_{\pi/2}^{3\pi/2} \phi_1(a, \theta) a(-\sin\,\theta)\, d\theta \\ &= \mathrm{i}\rho\omega \int_{\pi/2}^{3\pi/2} \phi_1^A(a, \theta) a(-\sin\,\theta)\, d\theta = \frac{-\pi\rho g H a}{2\cosh kh} c_1^A, \end{aligned} \tag{33}$$

$$\begin{aligned} F_z &= \mathrm{i}\rho\omega \int_{\pi/2}^{3\pi/2} \phi_1(a, \theta) a\, \cos\theta\, d\theta \\ &= \mathrm{i}\rho\omega \int_{\pi/2}^{3\pi/2} \phi_1^S(a, \theta) a\, \cos\theta\, d\theta \\ &= \frac{\rho g H a}{\cosh kh}\left[\sum_{n=1}^{M} \frac{2(-1)^n c_n^S}{4n^2 - 1} - 1\right], \end{aligned} \tag{34}$$

where Eqs. (31) and (32) have been used [21].

We note that for the solution in Section 3, the velocity potential ϕ_2 inside the breakwater will not decrease to zero when G approaches zero. At $G = 0$, the potential inside the breakwater satisfies the Laplace equation and the homogenous Neumann boundary condition at the close boundary. Then the general potential is not unique and equals to an arbitrary constant. If we use Eqs. (16)–(19) to calculate the expansion coefficients, the inner potential at $G = 0$ will be:

$$\phi_2 = \phi_2^S = \frac{-\mathrm{i}gH}{2\omega} \frac{1}{\cosh\, kh} d_0^S. \tag{35}$$

This is equivalent to adding a uniform pressure on the inside surface of the impermeable semi-circular arc and the bottom slab surface. Such a contribution is a matter of interpretation. As a result, when G approaches zero, the predicted vertical wave force on the semi-circular breakwater by Eq. (27) will not approach that by Eq. (34). So for the special case of an impermeable breakwater, the vertical wave force should be calculated by Eq. (34).

5. Numerical results

Linton and McIver [25] have discussed that the multipole expansion method usually converges fast with M. This has been confirmed in the present work for both the perforated case and the impermeable case. The numerical results of the reflection coefficient for an impermeable breakwater with different M at $a/h = 0.5$, 0.8 and 0.9 are listed in Tables 1–3, respectively. We note from these tables that for a semi-circular breakwater with smaller radius, the convergence of the solution is faster. Even for the case of large radius with $a/h = 0.9$, $M = 7$ is enough for obtaining sufficiently accurate results. Thus $M = 7$ is used in the present study.

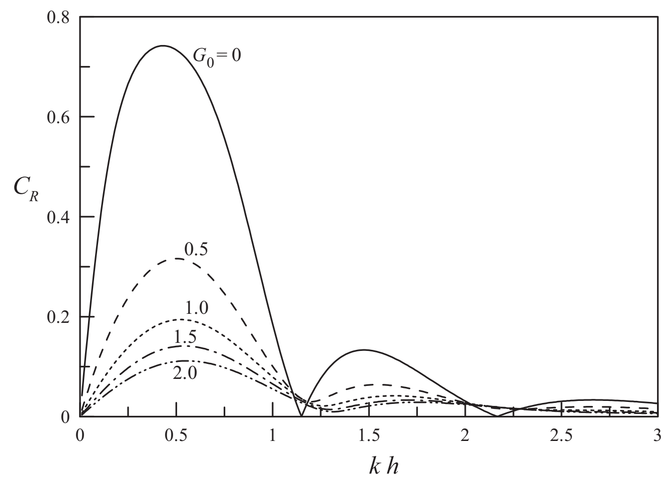

Fig. 2. Variation of the reflection coefficient C_R with kh at $a/h = 0.9$ and different G_0.

The reflection and transmission coefficients and the energy loss coefficient for a perforated semi-circular breakwater are examined in Figs. 2–4. According to Eq. (9), the porous effect parameter G may vary with the incident wave number k. Thus to plot curves of C_R, C_T and C_L as the function of kh, we use an alternative normalized porous effect parameter:

$$G_0 = \frac{h}{\delta} \frac{\varepsilon}{f - \mathrm{i}s} = Gkh. \tag{36}$$

We also noted that G is a complex quantity in theory and its real and imaginary parts denote, respectively, the resistance and inertial effects of the perforated plate on fluid motion [28]. But the inertial effect of the thin perforated plate may be not significant for the wave scattering problem in practice [30,31]. Thus we only consider a real G (G_0) in our analysis. It can be seen from Fig. 2 that the reflection coefficient decreases significantly with the increasing value of G_0. This is helpful to suppressing the seabed scour in front of the breakwater and the safe navigation of vessels near the structure. From Fig. 3, it can be seen that in comparison with an impermeable breakwater, a perforated breakwater may work well, resulting in a smaller transmission coefficient, which is a key point of the breakwater, besides absorbing wave energy. We also note from Fig. 3 that

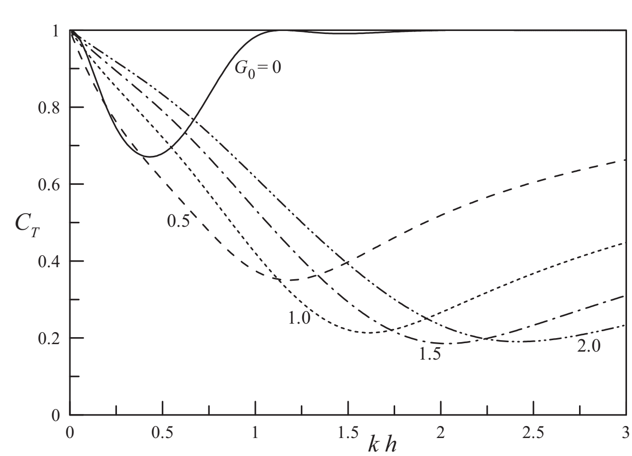

Fig. 3. Variation of the transmission coefficient C_T with kh at $a/h = 0.9$ and different G_0.

Table 1
Convergence of C_R with M at $a/h = 0.5$.

Truncated number	Reflection coefficient, C_R						
	$kh = 0.1$	$kh = 0.5$	$kh = 1.0$	$kh = 1.5$	$kh = 2.0$	$kh = 2.5$	$kh = 3.0$
$M = 1$	0.0490	0.2012	0.2345	0.1541	0.0623	0.0060	0.0170
2	0.0490	0.2015	0.2370	0.1622	0.0772	0.0257	0.0037
3	0.0490	0.2015	0.2370	0.1622	0.0773	0.0259	0.0043
4	0.0490	0.2015	0.2370	0.1622	0.0773	0.0259	0.0043
5	0.0490	0.2015	0.2370	0.1622	0.0773	0.0259	0.0043

Table 2
Convergence of C_R with M at $a/h = 0.8$.

Truncated number	Reflection coefficient, C_R						
	$kh = 0.1$	$kh = 0.5$	$kh = 1.0$	$kh = 1.5$	$kh = 2.0$	$kh = 2.5$	$kh = 3.0$
$M = 1$	0.2040	0.5547	0.2927	0.1156	0.2663	0.2740	0.2297
2	0.2084	0.5740	0.3539	0.0035	0.0913	0.0847	0.0789
3	0.2086	0.5746	0.3558	0.0108	0.0688	0.0390	0.0116
4	0.2086	0.5747	0.3559	0.0111	0.0678	0.0364	0.0059
5	0.2086	0.5747	0.3559	0.0111	0.0678	0.0363	0.0057
7	0.2086	0.5747	0.3559	0.0111	0.0678	0.0363	0.0057

Table 3
Convergence of C_R with M at $a/h = 0.9$.

Truncated number	Reflection coefficient, C_R						
	$kh = 0.1$	$kh = 0.5$	$kh = 1.0$	$kh = 1.5$	$kh = 2.0$	$kh = 2.5$	$kh = 3.0$
$M = 1$	0.3471	0.6767	0.0821	0.3763	0.4412	0.3965	0.3183
2	0.3759	0.7283	0.1754	0.1896	0.1716	0.1651	0.1947
3	0.3795	0.7328	0.1843	0.1408	0.0548	0.0111	0.0361
4	0.3799	0.7334	0.1853	0.1339	0.0344	0.0244	0.0161
5	0.3800	0.7335	0.1854	0.1330	0.0315	0.0297	0.0249
7	0.3800	0.7335	0.1854	0.1328	0.0310	0.0306	0.0264
9	0.3800	0.7335	0.1854	0.1328	0.0310	0.0306	0.0264

the minimum transmission coefficient occurs at larger kh with the increasing value of G_0. From Fig. 4, it is evident that for a fixed incident wave number, there is an optimum G_0 (corresponding to an optimum geometrical porosity of the plate) with maximum energy loss coefficient. In practice, the geometrical porosity of the breakwater should be carefully determined for obtaining optimum wave absorbing performance.

Figs. 5 and 6 give the variations of the dimensionless total horizontal and vertical wave forces, $C_{Fx} = |F_x|/(\rho gHa)$ and $C_{Fz} = |F_z|/(\rho gHa)$, for an impermeable breakwater with kh, respectively. As shown in Fig. 5, the value of C_{Fx} increases with kh, reaches its maximum and then decreases. The maximum value of C_{Fx} increases with a/h. We note from Fig. 6 that the value of C_{Fz} tends to unity when kh approaches zero. We also note from Fig. 6 that the variations of C_{Fz} versus kh are different at different values of a/h. For the cases of $a/h = 0.5$ and 0.8, C_{Fz} decreases monotonously with the increasing value of kh. However for a larger value of $a/h = 0.9$, C_{Fz} reaches a maximum value at a smaller value of $kh \approx 0.63$, and is zero at about $kh \approx 2.3$.

Fig. 7 gives the dimensionless phase difference C_θ (normalized by π) between the total horizontal and vertical wave forces on an impermeable breakwater at different a/h. We note that F_x and F_z are in-phase at $C_\theta = 0$, and out-phase at $C_\theta = 1$. For the case of out-phase,

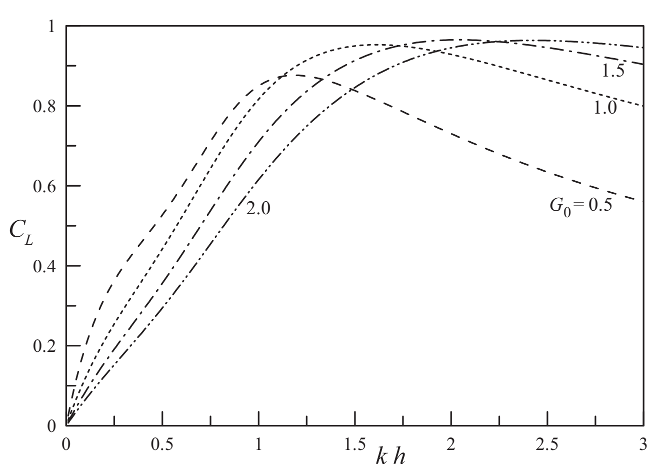

Fig. 4. Variation of the energy loss coefficient C_L with kh at $a/h = 0.9$ and different G_0.

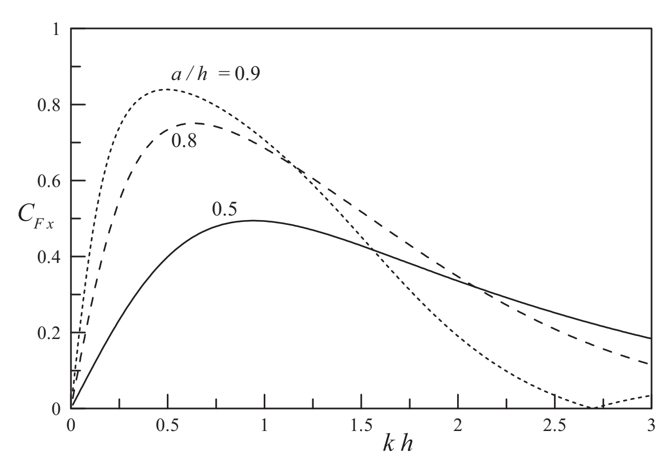

Fig. 5. Variation of the dimensionless total horizontal force C_{Fx} for an impermeable breakwater with kh at different a/h.

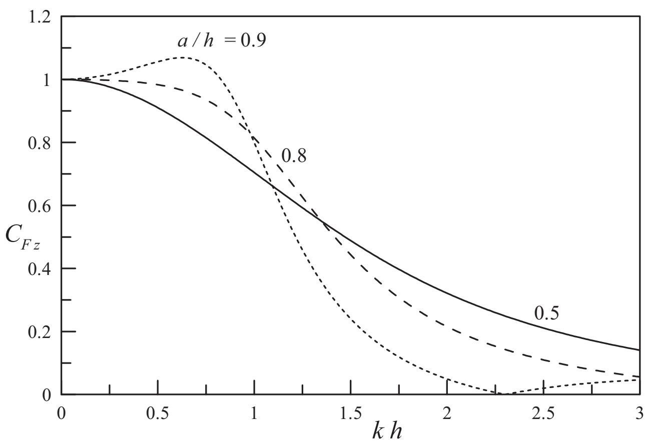

Fig. 6. Variation of the dimensionless total vertical force C_{Fz} for an impermeable breakwater with kh at different a/h.

the vertical wave force corresponding to the maximum horizontal force also attains maximum and directs downwards. This is most helpful to enhancing the stability against sliding of the breakwater. By comparing Fig. 7 with Figs. 5 and 6, we note that when the total horizontal and vertical wave forces attain large values, the phase difference between them is also large. Then the large phase difference is helpful to the stability against sliding of the submerged semi-circular breakwater.

Now we give Figs. 8–10 to examine the C_{Fx}, C_{Fz} and C_θ for a perforated semi-circular breakwater. The value of $a/h = 0.9$ and different G_0 are used in these figures. We note from Fig. 8 that the maximum value of C_{Fx} decreases significantly with the increasing value of G_0. This is beneficial to enhancing the stability against sliding of the breakwater. From Fig. 9, we note that when the semi-circular breakwater is perforated, the total vertical force decreases significantly at $kh < 1.5$. Moreover, we note from Fig. 10 that in comparison with the impermeable breakwater, the phase difference between total horizontal and vertical wave forces on the perforated breakwater increases significantly. This is also helpful to enhancing the stability against sliding of the breakwater as discussed in Fig. 7.

For the perforated semi-circular breakwater, the total vertical wave force includes two components acting on the semi-circular arc and the bottom slab. The variations of these two dimensionless component forces, $C_{Fz1} = |F_{z1}|/(\rho gHa)$ and $C_{Fz2} = |F_{z2}|/(\rho gHa)$, with kh at $a/h = 0.9$ and different G_0 are plotted in Figs. 11 and 12, respectively. We note from Fig. 11 that the value of C_{Fz1} decreases with the increasing value of G_0. This is natural in physics. We also note from Fig. 11 that when kh approaches zero, C_{Fz1} also approaches zero. This is rather different from the impermeable semi-circular breakwater, where the value of C_{Fz} (= C_{Fz1}) approaches unity when kh approaching zero (See Fig. 6). From Fig. 12, we can see that the value of C_{Fz2} increases with the increasing value of G_0 at the moderate value of $kh \approx 0.8$–2.0. When kh decreases to zero, the values

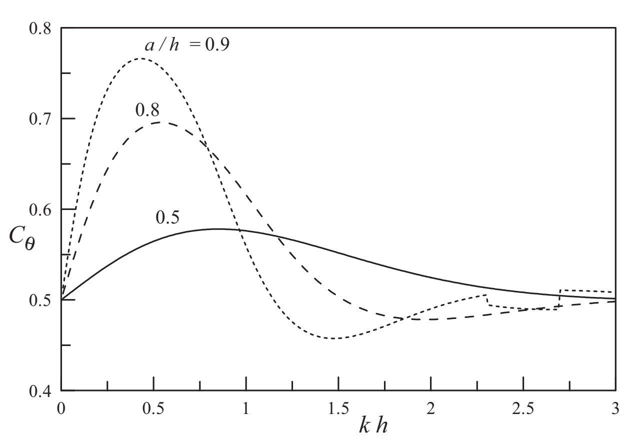

Fig. 7. Variation of the dimensionless phase difference C_θ between F_x and F_z for an impermeable breakwater with kh at different a/h.

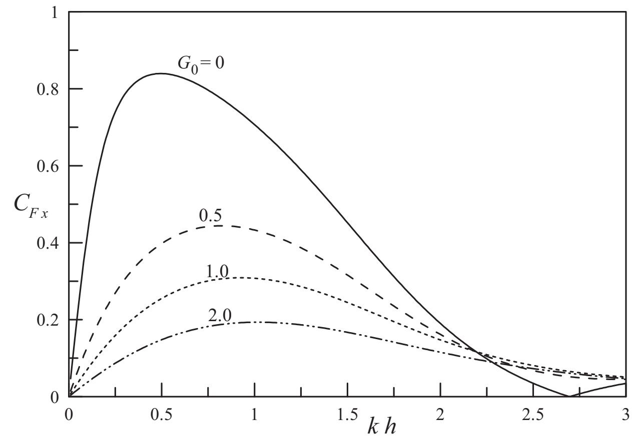

Fig. 8. Variation of the dimensionless total horizontal force C_{Fx} with kh at $a/h = 0.9$ and different G_0.

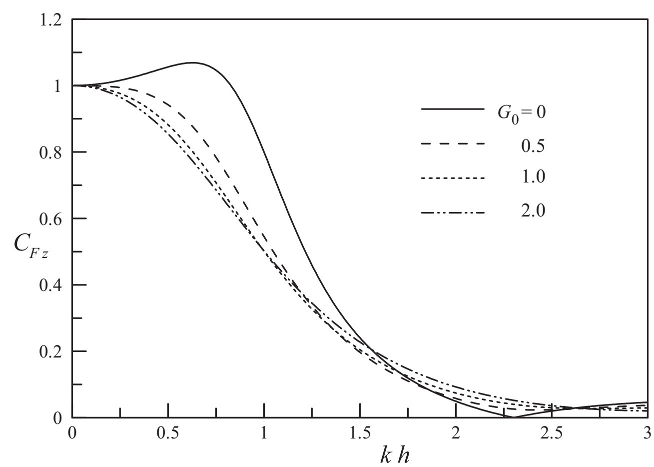

Fig. 9. Variation of the dimensionless total vertical force C_{Fz} with kh at $a/h = 0.9$ and different G_0.

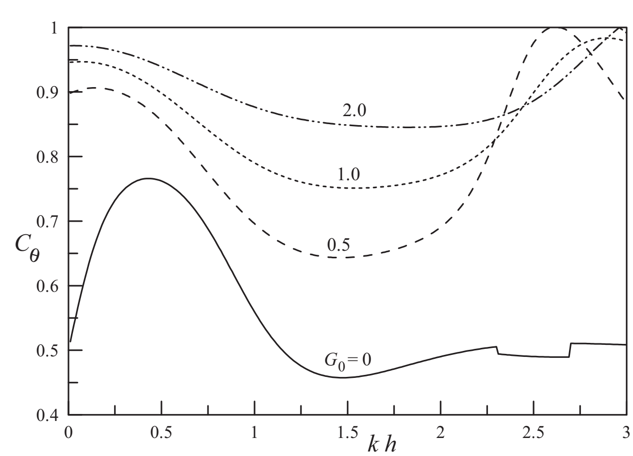

Fig. 10. Variation of the dimensionless phase difference C_θ between F_x and F_z with kh at $a/h = 0.9$ and different G_0.

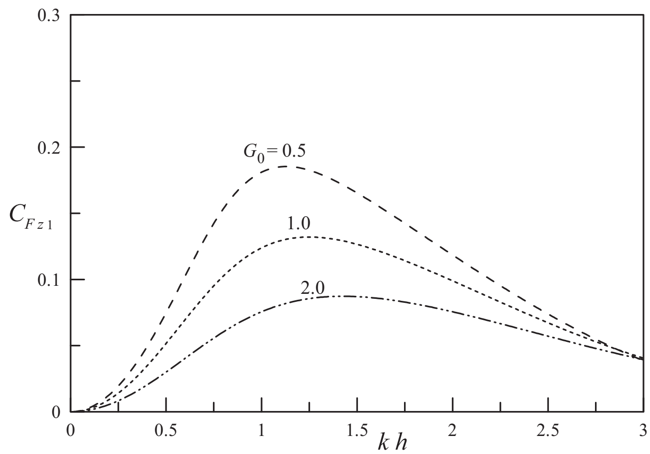

Fig. 11. Variation of the dimensionless vertical force on the semi-circular arc C_{Fz1} with kh at $a/h = 0.9$ and different G_0.

of C_{Fz2} at different G_0 all approaches unity. If there is no the perforated semi-circular arc (G_0 = infinity), the dimensionless vertical wave force acting on the bottom slab is

$$C_{Fz2} = \left| \frac{1}{\cosh\ kh} \frac{\sin ka}{ka} \right|. \tag{37}$$

Obviously the limit of C_{Fz2} is unity at $kh = 0$. We note that adding a modification coefficient including the effect of the perforated semi-circular arc into Eq. (37) may give a practical formula for calculating the vertical wave force acting on the bottom slab.

Apart from the wave forces, one may have an interest in the pressure distribution along the semi-circular breakwater in practice. So the normalized pressure, $C_p = |p|/(\rho gH)$, acting on an impermeable semi-circular breakwater is plotted in Fig. 13 at $a/h = 0.9$ and $kh = 0.5$, 1.0, 1.5 and 2.5. We note from Fig. 13 that the maximum pressure generally occurs at the seaside top area of the semi-circular arc. So the structural strength of this area should be carefully considered in practice. Also it is interesting to find in Fig. 13 that although the total wave forces at $kh = 1.0$ are smaller than that at $kh = 0.5$ (see Figs. 5 and 6), the maximum pressure at $kh = 1.0$ is larger than that at $kh = 0.5$.

For a perforated semi-circular breakwater, the distribution of the normalized pressure difference, $C_p = |p_2 - p_1|/(\rho gH)$, between the two sides of the perforated arc is plotted in Fig. 14. Here the parameters used in the calculations are: $G_0 = 1.0$, $a/h = 0.9$ and $kh = 0.5$, 1.0, 1.5 and 2.5. At $kh = 1.0$, 1.5 and 2.5, the distribution of C_p is similar to that of the impermeable breakwater given in Fig. 13 although the maximum value of C_p is reduced. But at $kh = 0.5$, the distribution of C_p is rather different from the impermeable breakwater and the minimum pressure difference occurs at the leeside top area of the semi-circular arc. Fig. 15 gives the distribution of

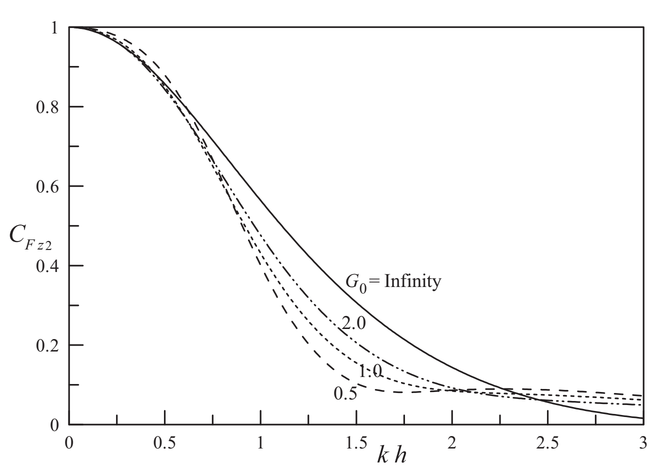

Fig. 12. Variation of the dimensionless vertical force on the bottom slab C_{Fz2} with kh at $a/h = 0.9$ and different G_0.

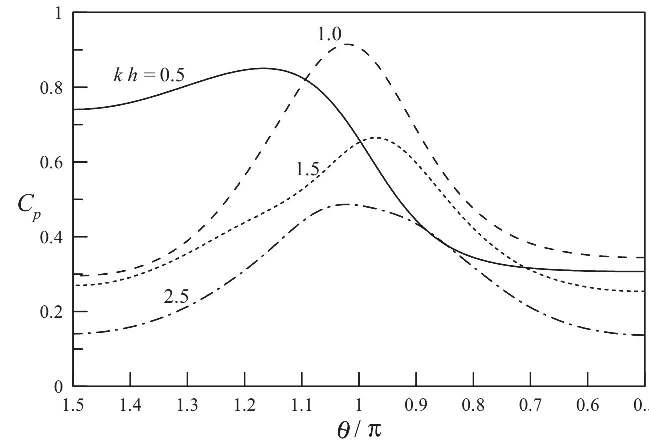

Fig. 13. Distribution of the normalized pressure difference C_p over an impermeable breakwater at different kh.

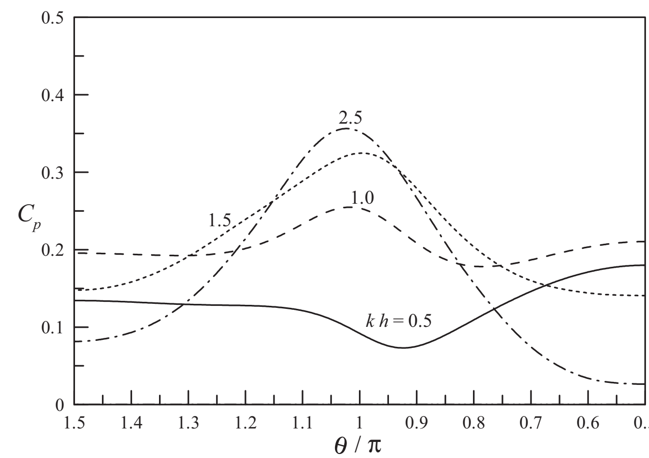

Fig. 14. Distribution of the normalized pressure difference C_p over the perforated arc at $G_0 = 1.0$, $a/h = 0.9$ and different kh.

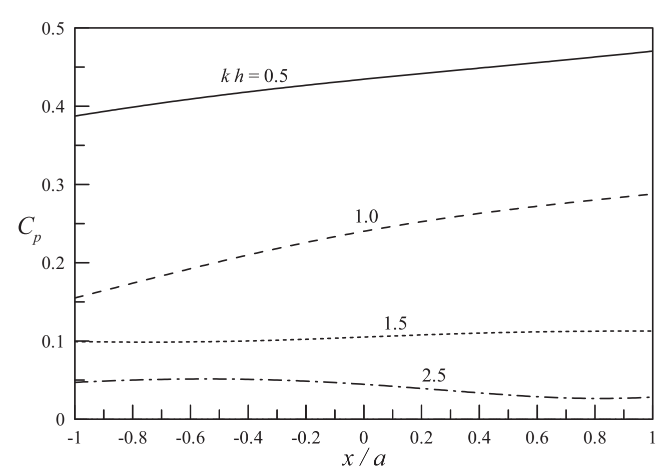

Fig. 15. Distribution of the normalized pressure C_p over the bottom slab at $G_0 = 1.0$, $a/h = 0.9$ and different kh.

the normalized pressure, $C_p = |p_2|/(\rho gH)$, on the bottom slab of a perforated semi-circular breakwater. Here the parameters used in the calculations are the same as that in Fig. 14. It can be seen from this figure that the variation of the pressure along the bottom slab is not acute. At $kh = 0.5$ and 1.0, the larger value of C_p occurs at the leeside. However this is just the opposite at $kh = 2.5$.

6. Concluding remarks

On the basis of the linear potential theory, we have developed multipoles with singularities on the seabed and used the multipole expansions to obtain a solution of water wave interaction with submerged perforated semi-circular breakwaters. The convergence of the solution is fast. The hydrodynamic performance of submerged semi-circular breakwaters has been carefully examined. A submerged perforated semi-circular breakwater with suitable porosity may work well reducing the transmission and reflection coefficients. Perforating the semi-circular arc can significantly reduce the total horizontal and vertical wave forces acting on the breakwater. The maximum total horizontal and vertical wave forces acting on a perforated or impermeable semi-circular breakwater has a significant phase difference, which is beneficial to enhancing the stability against sliding. The maximum wave pressure difference on the semi-circular arc generally occurs at the seaside top area of the arc.

Acknowledgements

We wish to thank Prof. Guo-Xiong Wu at UCL for his most valuable private notes on this work. Without his collaboration, this paper could not have been finished. This work was supported by the National Natural Science Foundation of China (Grant Nos. 51010009 and 50909086) and the Natural Science Foundation of Shandong Province, China (Grant No. Q2008F01). We would like to thank the referees for their valuable comments, which enhanced the quality of this paper.

Appendix A. Multipoles with singularities on the seabed

Here we give symmetric and antisymmetric multipoles, which are singular at the centre of the circle $(0, -h)$ and satisfy Eqs. (2)–(5). The procedure is similar to that used in the references discussed in the introduction.

The sketch of the Cartesian and polar coordinate systems are shown in Fig. 1, where $r\cos\theta = -(z+h)$ and $r\sin\theta = x$. The symmetric and antisymmetric multipole expansions for the present problem, which are singular at $(0, -h)$ and satisfy seabed condition in Eq. (4), can be written, respectively, as:

$$\tilde{\varphi}_n^S = \frac{\cos 2n\theta}{r^{2n}}, \quad n = 1, 2, \ldots, \tag{A.1}$$

$$\tilde{\varphi}_n^A = \frac{\sin(2n-1)\theta}{r^{(2n-1)}}, \quad n = 1, 2, \ldots \tag{A.2}$$

From Gradshteyn and Ryzhik ([32], Eqs. 3.944(5) and (6)), we have:

$$\frac{\cos 2n\theta}{r^{2n}} = \frac{1}{(2n-1)!}\int_0^\infty \mu^{(2n-1)} e^{-\mu(z+h)} \cos\mu x\, d\mu, \quad z > -h, \tag{A.3}$$

$$\frac{\sin(2n-1)\theta}{r^{(2n-1)}} = \frac{1}{(2n-2)!}\int_0^\infty \mu^{(2n-2)} e^{-\mu(z+h)} \sin\mu x\, d\mu, \quad z > -h. \tag{A.4}$$

The antisymmetric multipoles may be written as:

$$\varphi_n^A = \tilde{\varphi}_n^A + H_n^A, \tag{A.5}$$

where H_n^A is to be determined through Eqs. (2)–(5).

The expression of H_n^A satisfying Eq. (4) can be written as:

$$H_n^A = \frac{1}{(2n-2)!}\int_0^\infty f(\mu)\mu^{(2n-2)} \cosh\mu(z+h)\sin\mu x\, d\mu. \tag{A.6}$$

Inserting $\tilde{\varphi}_n^A + H_n^A$ into Eq. (3), we have that:

$$f(\mu) = \frac{(\mu+K)e^{-\mu h}}{\mu\sinh\mu h - K\cosh\mu h}. \tag{A.7}$$

Then

$$H_n^A = \frac{1}{(2n-2)!}\int_0^\infty \frac{(\mu+K)\mu^{(2n-2)}e^{-\mu h}\cosh\mu(z+h)}{\mu\sinh\mu h - K\cosh\mu h}\sin\mu x\, d\mu, \tag{A.8}$$

where the path of the integration passes below the point at $\mu = k$ for satisfying the radiation condition in Eq. (5) [33].

Now the antisymmetric multipoles on the seabed can be written as:

$$\varphi_n^A = \frac{\sin(2n-1)\theta}{r^{(2n-1)}} + \frac{1}{(2n-2)!}\int_0^\infty \frac{(\mu+K)\mu^{(2n-2)}e^{-\mu h}\cosh\mu(z+h)}{\mu\sinh\mu h - K\cosh\mu h}\sin\mu x\, d\mu, \tag{A.9}$$

where the path of the integration passes below the point at $\mu = k$. As $x \to \pm\infty$, we have:

$$\varphi_n^A \sim \pm\frac{\pi k^{(2n-2)}\cosh k(z+h)}{2hN_0^2(2n-2)!}e^{\pm ikx}, \tag{A.10}$$

where, $N_0^2 = 0.5[1+\sinh 2kh/(2kh)]$. Using the same method as above, we can obtain the symmetric multipoles on the seabed:

$$\varphi_n^S = \frac{\cos 2n\theta}{r^{2n}} + \frac{1}{(2n-1)!}\int_0^\infty \frac{(\mu+K)\mu^{(2n-1)}e^{-\mu h}\cosh\mu(z+h)}{\mu\sinh\mu h - K\cosh\mu h}\cos\mu x\, d\mu, \tag{A.11}$$

where the path of the integration passes below the point at $\mu = k$. As $x \to \pm\infty$, we have:

$$\varphi_n^S \sim \frac{\pi i k^{(2n-1)}\cosh k(z+h)}{2hN_0^2(2n-1)!}e^{\pm ikx}. \tag{A.12}$$

The multipoles can be further written as power series expansions:

$$\varphi_n^S = \frac{\cos 2n\theta}{r^{2n}} + \sum_{m=0}^{\infty} C_{mn}^S r^{2m}\cos 2m\theta, \tag{A.13}$$

$$\varphi_n^A = \frac{\sin(2n-1)\theta}{r^{(2n-1)}} + \sum_{m=1}^{\infty} C_{mn}^A r^{(2m-1)}\sin(2m-1)\theta, \tag{A.14}$$

where

$$C_{mn}^S = \frac{1}{(2m)!(2n-1)!}\int_0^\infty \frac{(\mu+K)\mu^{(2m+2n-1)}e^{-\mu h}}{\mu\sinh\mu h - K\cosh\mu h}d\mu, \tag{A.15}$$

$$C_{mn}^A = \frac{1}{(2m-1)!(2n-2)!}\int_0^\infty \frac{(\mu+K)\mu^{(2m+2n-3)}e^{-\mu h}}{\mu\sinh\mu h - K\cosh\mu h}d\mu, \tag{A.16}$$

in which the path of the integration passes below the point at $\mu = k$.

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